SOC L1 LAB · INC-006

INC-006: Authorized Change Triage

Four persistence alerts, one approved change, no incident

INCIDENT	INC-006, Authorized change triage
REPORT DATE	2026-11-01
CLASSIFICATION	Lab exercise, sanitized for publication
ENVIRONMENT	Wazuh 4.14.7 · Debian 12 · QEMU/KVM
SEVERITY	None (authorized change)

Independent portfolio lab. No production systems, client data, or real credentials.

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Case	INC-006, Authorized change triage
Classification	Lab exercise, sanitized for publication
Activity date	2026-10-31 / 2026-11-01
Report date	2026-11-01
Author	SOC L1 portfolio lab
Decision	Closed, authorized change, no incident
Severity	None (the alerts were correct. The activity was approved)
Related	`tickets/INC-006-authorized-change.md` , `docs/evidence/INC-006-*`

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1. Executive summary

At 23:35:51 UTC, four persistence alerts fired:

Rule	Path
100129	`/etc/cron.d/log-maintenance`
100130	`/etc/systemd/system/logrotate-check.service`
100130	`/etc/systemd/system/logrotate-check.timer`
100130	`/etc/systemd/system/timers.target.wants/logrotate-check.timer`

They matched an approved change, CHG-2026-1101-01, requested by `analyst` and approved by the lab change manager. The change record listed the three files and their SHA-256 values. Every deployed file matched. The timer was active. The cron job was a benign log-cleanup task. No authentication, process, or network anomaly followed.

Decision: closed as an authorized change, not an incident

This is not a false positive. The alerts correctly reported that files changed. The context, an approved change with matching hashes, made the activity benign. Calling it a false positive would mislabel a working detection and hide the fact that the control worked.

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2. Scope and method

2.1 Scope

In scope	Out of scope
One Debian 12 VM (`soc-victim` , 10.0.2.15)	Production systems
Change CHG-2026-1101-01	Real business changes
Rules 100129/100130	New detection rules
Evidence verification	Exploitation

2.2 Method

1. Write a change record with scope, files, hashes, approver, and rollback.
2. Deploy the cron file and systemd service/timer exactly as recorded.
3. Collect the four alerts.
4. Verify the change record, hashes, file contents, timer state, and absence of follow-on anomalies.
5. Close the case with a written rationale.

2.3 Limitations

- The change manager and approval are synthetic.
- One host. No CMDB or ticketing integration.
- The cron job runs daily at 02:15. It did not execute during the 15-minute triage window. The timer is scheduled daily at midnight.

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3. The change

CHG-2026-1101-01, Nightly log maintenance

Field	Value
Status	Approved
Requester	analyst
Approver	lab-change-manager
Window	2026-11-01 06:30 UTC
Purpose	Remove rotated logs older than 7 days. Add a logrotate health check
Risk	Low

Files and hashes:

Path	SHA-256
`/etc/cron.d/log-maintenance`	`58acf8face227878d58765d32cebe4b82b562996b9e77b02d7011e7533835437`
`/etc/systemd/system/logrotate-check.service`	`6a0555d9e6831f2dcf2b208e7f7bf4da9473f145b4d31b46ebde76def3a50f19`
`/etc/systemd/system/logrotate-check.timer`	`83d683d881cc93e29c2f56f6925756b60d53c84f4234e2bbdeac060d87658cd8`

The cron file:

```
# Nightly cleanup of rotated logs older than 7 days
15 2 * * * root /usr/bin/find /var/log -type f -name '*.gz' -mtime +7 -delete
```

The service runs `/usr/bin/true`. The timer triggers it daily. Neither does anything destructive.

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4. Timeline

Time (UTC)	Event
23:35:45	Triage window starts. Change record staged
23:35:51	Cron file, service, and timer written
23:35:51.674	100129 `/etc/cron.d/log-maintenance` added
23:35:51.685	100130 `logrotate-check.service` added
23:35:51.692	100130 `logrotate-check.timer` added
23:35:51.883	100130 enable symlink added
23:35:52	Timer active
23:36	Hashes verified against the change record
23:36	No authentication or process anomalies
23:37	Case closed as authorized change

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5. Alerts

Rule	Level	Path	Assessment
100129	12	`/etc/cron.d/log-maintenance`	Expected by CHG
100130	10	`logrotate-check.service`	Expected by CHG
100130	10	`logrotate-check.timer`	Expected by CHG
100130	10	`timers.target.wants/...`	Expected side effect of `enable`

Supporting evidence:

- audit `80792` `tee` / `cp` commands into the three paths.
- sudo `5402` attribution to `analyst`.
- FIM 554 events for the change record under `/home/analyst/changes`.
- `systemctl cat` matching the recorded content.
- `systemctl list-timers` showing the next run.
- no failed logins, invalid users, or authentication failures in the window.

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6. Investigation

6.1 Authorization

A change record exists, names the requester and approver, lists the exact files, and gives the rollback. The deployment time falls inside the approved window.

6.2 Integrity

All three deployed files match the hashes in the change record. No extra file was written under `/etc/cron.d` or `/etc/systemd/system` during the window.

6.3 Behaviour

- The cron job deletes only `*.gz` files older than seven days under `/var/log`.
- The service runs `/usr/bin/true`. It has no side effect.
- The timer is enabled for daily execution and has not run during triage.

6.4 Follow-on activity

No authentication failures, new accounts, unusual processes, or outbound connections appeared after the change. The host's normal telemetry continued.

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7. Why this is not a false positive

A false positive is an alert that should not have fired: the rule misread the event, or the event did not happen.

Here, the rule read the event correctly:

- a cron file did appear.
- a service and timer did appear.
- an enable symlink did appear.

The analyst's job was not to dispute the alert. It was to establish whether the change was authorized. The change record and matching hashes did that.

The correct label is benign true positive or authorized change. The distinction matters:

Term	Meaning	Action
False positive	The detection was wrong	Fix the rule
Benign true positive	The detection was right. The activity is allowed	Close with evidence. Tune context
True positive	The detection was right. The activity is malicious	Incident response

Closing this as a "false positive" would have created pressure to suppress 100129/100130, which would hide the next unauthorized cron file.

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8. Decision

Closed, authorized change, no incident

- No containment.
- No rule change.
- Change record retained.
- Hashes and alert set preserved.
- The detection remains enabled.

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9. What would change the decision

Condition	Outcome
No change record	Escalate as unapproved persistence
Hash mismatch	Escalate. The deployed file is not the approved one
Deployment outside the window	Escalate. Ask the approver
Unexpected command in the cron/service	Escalate. The content is not the approved content
Authentication failure or new account	Escalate. Possible compromise
Outbound connection after the change	Escalate. Possible payload

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10. Recommendations

#	Action
1	Enrich 100129/100130 with the change ID when one exists
2	Route approved changes to a queue that expects them
3	Keep the rules enabled. Do not suppress on busy change days
4	Store file hashes in the change record
5	Add an allow-list of approved unit names, not a blanket suppression
6	Review the change after the first scheduled run

11. Lessons learned

- **A correct alert is not a false positive**
  Context decides benign vs malicious.
- **Change control is the control**
  FIM proved the files appeared. The change record proved they were allowed.
- **Hashes make the decision auditable**
  "Looks fine" is not evidence.
- **Do not suppress the rule**
  Suppression would hide the next unauthorized change.
- **Write the "no incident" ticket as carefully as the incident ticket**
  It is the hardest L1 skill to demonstrate.

Appendix A: Change record

```
# CHG-2026-1101-01: Nightly log maintenance

| Field | Value |
|---|---|
| Status | Approved |
| Requester | analyst |
| Approver | lab-change-manager |
| Window | 2026-11-01 06:30 UTC |
| Purpose | Remove rotated logs older than 7 days; add a logrotate health check |
| Risk | Low |

Files and SHA-256 values are recorded in the change record and verified above.
Rollback: disable the timer, remove the units and cron file, daemon-reload.
```

Appendix B: MITRE ATT&CK

The alerts map to persistence techniques:

Technique	ID	Note
Scheduled task/job: cron	T1053.003	Authorized maintenance
Create/modify system process: systemd	T1543.002	Authorized timer

The mapping is why the rules exist. It does not make an approved change malicious.

Appendix C: Evidence index

File	Contents
`docs/evidence/INC-006-alerts.json`	162 alerts in the triage window
`docs/evidence/INC-006-change-record.md`	Approved change with hashes
`docs/evidence/INC-006-triage.txt`	Unit contents, hashes, timer state
`tickets/INC-006-authorized-change.md`	The closure decision

Appendix D: Glossary

Term	Meaning
Benign true positive	Correct alert, allowed activity
False positive	Incorrect alert
Change record	Documented approval, scope, hashes, rollback
FIM	File integrity monitoring